



Speeding up the development of solid state electrolyte by machine learning

Qianyu Hu^a, Kunfeng Chen^{a,*}, Jinyu Li^b, Tingting Zhao^{b,*}, Feng Liang^{c,*}, Dongfeng Xue^{d,*}

^a Institute of Novel Semiconductors, State Key Laboratory of Crystal Materials, Shandong University, Jinan 250100, China

^b College of Artificial Intelligence, Tianjin University of Science & Technology, Tianjin 300457, China

^c Key Laboratory for Nonferrous Vacuum Metallurgy of Yunnan Province, National Engineering Research Center of Vacuum Metallurgy, Faculty of Metallurgical and Energy Engineering Kunming University of Science and Technology, Kunming 650093, China

^d Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, Shenzhen 518110, China



ARTICLE INFO

Keywords:

Solid state electrolyte
Machine learning
Machine learning model
Database

ABSTRACT

Solid-state electrolytes have been demonstrated immense potential with their high density and safety for Li, Na batteries. The discovery of novel crystals is of fundamental scientific and technological interest in solid-state chemistry. The discovery, synthesis and application of energetically favourable solid-state electrolytes has been bottlenecked by expensive trial-and-error approaches. Machine learning has brought breakthroughs to solid-state electrolytes. Numerous solid-state electrolyte candidates have been screened by different models at multiscale, i.e., interatomic potentials, molecular dynamics, ionic conductivity. Machine learning method also accelerate the synthesis prediction, mechanism discovery and interface design. This review would answer the question what can be done for solid-state electrolytes by machine learning, including descriptor, model, algorithm etc. This paper will promote fast integration between scientists in materials, software, computing discipline.

1. Introduction

The development of commercial portable electronic products, the rapid transition from fossil-fuel-based automobiles to electric vehicles, and the optimization of smart grids have significantly spurred the growth in demand for commercial rechargeable batteries. These advancements impose more stringent requirements on electrolyte materials—a key component in batteries that controls overall electrochemical performance [1,2]. Currently, commercial liquid electrolytes are limited in their future development due to issues such as low stability, a limited electrochemical window, low energy density, and high cost [3]. In particular, the propensity of liquid electrolytes to leak and their flammability present significant safety concerns for commercialization [4] and may even pose threats to human safety [5]. As a viable alternative, solid-state electrolytes (SSE) demonstrate immense potential with their high density and safety. The inherent non-flammability of SSEs allows for safer and more efficient operation at high temperatures, thereby circumventing the safety issues associated with liquid electrolytes. Moreover, SSEs possess higher mechanical strength, which can inhibit the formation of lithium dendrites, and have a high electrochemical window of over 4 V, enabling higher energy densities. Their superior electrochemical stability may allow the use of electrodes with higher specific capacities [6,7].

SSEs come in various types, including halides, oxides, sulfides, and polymers. Despite their significant advantages and widespread attention from the academic and industrial communities, SSEs have not yet been widely adopted. The main reason is that the balance of performance depends on various parameters, and many known SSEs are limited by one or several key properties, such as relatively low ionic conductivity ($\sigma < 10^{-3}$ S/cm), shorter cycle life (< 200 cycles), or poor interface compatibility (poor mechanical properties, mismatched coefficients of thermal expansion, etc) [8–11]. For example, sulfide-based SSEs can reach ionic conductivity levels comparable to liquid electrolytes, but they are prone to side reactions that increase interfacial charge transfer resistance, thereby reducing battery performance [12]. Polymer SSEs, such as polyethylene oxide, though flexible and easy to manufacture, usually exhibit lower ionic conductivity [13]. Mechanical damage can also lead to a decline in electrolyte performance and interfacial dynamics [14]. Therefore, a significant amount of work is currently focused on developing and designing SSE materials with multiple desirable properties, optimizing electrolyte interfaces, and conducting mechanistic analysis and structural optimization [15–20].

Typically, experimental, and theoretical work are the pillars supporting materials science research, with experimental studies demanding significant resources and equipment. In addition, material development often relies on human experience, intuition, and even

* Corresponding authors.

E-mail addresses: kunfeng.chen@sdu.edu.cn (K. Chen), tingting@tust.edu.cn (T. Zhao), liangfeng@kust.edu.cn (F. Liang), dfxue@uestc.edu.cn (D. Xue).

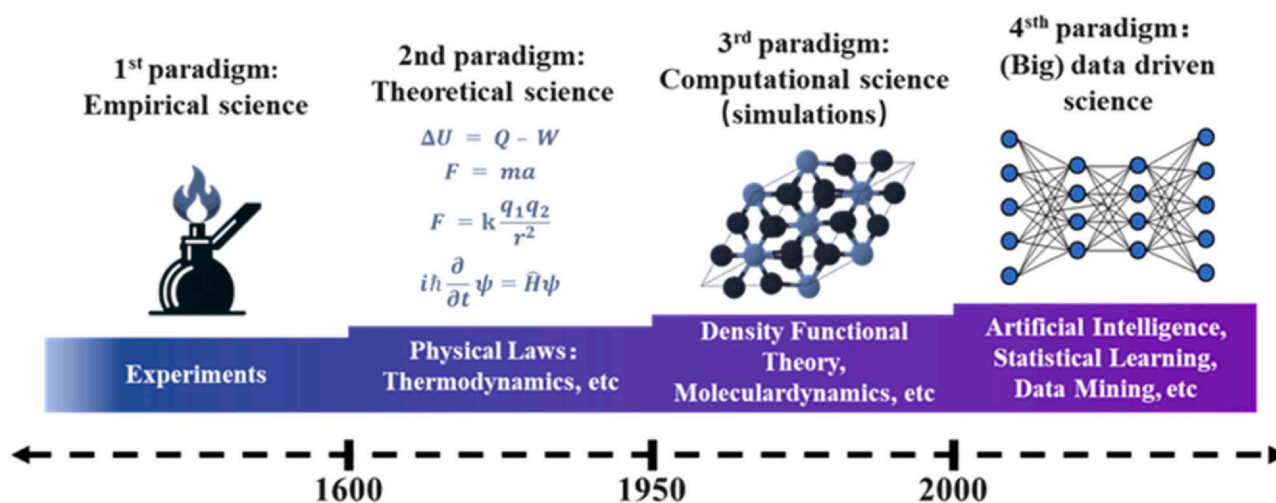


Fig. 1. The four paradigms of science: empirical, theoretical, computational, and data driven. Adapted with the permission from Ref. [23], Copyright 2016, AIP Publishing.

serendipity. The development of a material often remains challenged in commerce until it is integrated into new technology, due to various cost constraints [21]. Despite the growing demand for materials, acquiring superior materials remains limited by high resource consumption, inefficiency, and high costs.

The advent of computational simulation methods has propelled the rapid development of materials science. Techniques such as density functional theory (DFT), molecular dynamics, and Monte Carlo simulations can model complex real-world problems, avoiding human and experimental limitations. However, these methods are computationally demanding, time-consuming, and resource-intensive [22]. In recent years, with increased computational power and an abundance of experimental data, the application of machine learning (ML) as a new tool in materials science has become feasible. Combining big data and artificial intelligence, it is referred to as the "fourth paradigm of science" [23] (Fig. 1). As a branch of artificial intelligence, ML aims to identify potential mapping relationships by analyzing historical data and circumstances, significantly shortening the cycle of material design and discovery. This data-driven scientific approach naturally integrates methods of data acquisition from experiments, theory, and computational simulations, and processes tasks through feature handling, model training, and analytical optimization to obtain results [24] (Fig. 2).

In addressing the current issues related to SSEs, the integration of ML represents a positive trend and has already yielded many achievements. We believe that the application of ML in the field of SSEs has made significant progress, but continued efforts are necessary. This article reviews past scholars' ML applications in SSEs, starting from exploring the construction of the SSEs database, discussing the application methods of ML, and finally discussing our vision for future research in this field.

2. What can be done for SSE by ML?

The core of ML relies on statistical and optimization algorithms. It optimizes scientific models by analyzing fundamental patterns and relationships within data, thus enabling new predictions and decisions [25]. For instance, in traditional computational approaches, selecting high ionic conductivity electrolytes involves considering various related material properties and attempting to list complex equations for solving, which is highly challenging. In contrast, ML methods try to infer the mapping between ionic conductivity and SSE systems by observing data from different types of SSEs, deducing potential association patterns to predict high ionic conductivity SSEs. This statistical method of thinking can be applied to various aspects of SSEs.

Many of the statistical methods used in ML have existed for decades, but their application in materials science is a recent development. Specifically, in the study of SSE materials, the earliest applications of ML date back to around 2010. At that time, the application of ML was limited by technology and available open-source tools. Early research, such as the work by Xu et al. [26], used experimental and literature data to establish ML models to predict the properties of perovskite electrolytes, combined with first-principles calculations. That study mainly focused on improving model accuracy but did not consider other relevant properties of SSEs, which, from today's perspective, is undoubtedly one-sided.

Since the introduction of the "Materials Genome Initiative" by the United States in 2011, the demand for high-quality materials databases and the increase in publicly accessible databases have led to rapid development of ML in materials science [27]. With the advancement of automated high-throughput technologies, generating large first-principles datasets has become possible, although experimental data and

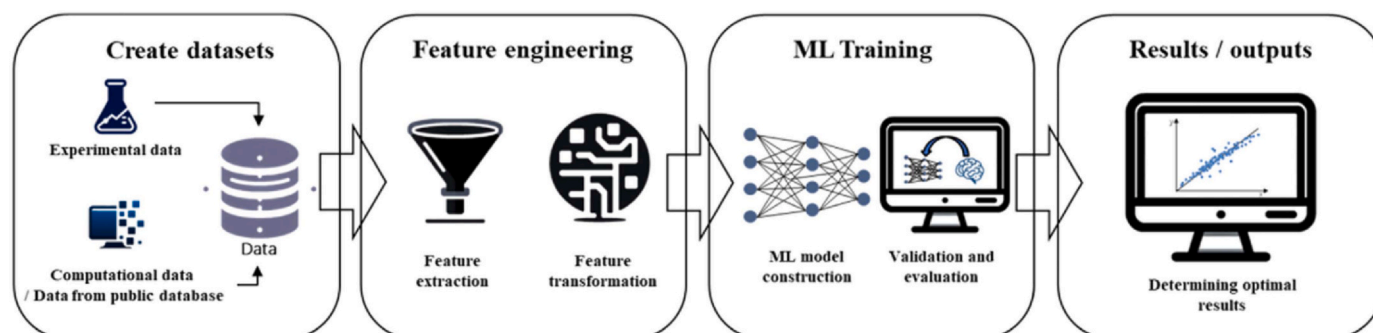


Fig. 2. A workflow for constructing ML models.

theoretical calculation data still dominate datasets in the short term. This has greatly advanced the application of ML in materials science, and many relevant reviews have emerged. For example, automatic feedback loops can be used to accelerate material synthesis and device manufacturing, accelerate the evaluation of material properties and the establishment of useful data sets, determine the material structure and discover new materials, and automatically discover new materials science Laws and principles, etc[28–32].

From 2012–2018, applications of ML in the field of SSEs were relatively few. However, since 2019, research outputs have been increasing year by year, showing that ML is gradually changing the paradigm of SSE research [33–39]. Currently, applications of ML in the field of SSEs can be divided into five main areas: 1) Optimizing computational simulation methods; 2) Predicting the properties of SSEs; 3) Guiding the synthesis of new SSEs; 4) Optimizing experimental parameters; 5) Upgrading characterization methods. These applications demonstrate the immense potential of ML in enhancing the efficiency and depth of research in SSEs.

2.1. Optimizing Computational Simulation Methods

The structure and composition of SSE materials are complex, and relying solely on molecular simulation and quantum mechanics-based computational methods can be limiting, with their time scales being daunting. Combining ML methods with traditional computational methods like density functional theory to accelerate material design significantly speeds up the computational process and is considered a viable approach for predicting electrolytes [40]. Interestingly, this optimization approach is flexible. For example, Fujimura et al. [41] combined ML with DFT to predict lithium-ion conductivity values for 72 components at 373 K (Fig. 3A). They used a support vector regression (SVR) model in conjunction with first-principles calculations to expedite the computational process. Sendek et al. [42] combined DFT-MD simulations with logistic regression models to screen over 12,000 candidate compounds of different structures and compositions. Their model was more effective at identifying superionic conductors than random guessing. Fig. 3B shows the calculated ionic conductivities of the 10 conductors predicted by the ML model. Jalem et al. compared two ML methods, Partial Least Squares (PLS) regression [43] and Artificial Neural Network (ANN) models [44], to enhance DFT calculations.

Traditional molecular dynamics (MD) simulations rely on classical potential energy functions to describe interactions between atoms, which are usually simplified models based on physical principles. ML potentials learn and predict these interactions through ML models, rather than directly relying on traditional potential functions. ML models learn the relationship between atomic configurations and their corresponding potential energy, and trained ML models can directly predict the potential energy of atomic structures with given Cartesian coordinates [45]. This is a promising approach to solving computationally challenging problems. Lacivita et al. [46] used a combination of ML potentials and genetic algorithm (GA) to determine doped near-ground state crystal structures and used these structures to estimate the defect formation energy for N insertion into Li_3PO_4 .

ML potential simulations can also address issues like the diffusion characteristics of Li in complex SSE systems [47] (Fig. 3C), predicting lithium conductivity and activation barriers [48], and gaining insights into the effects of doping on the diffusion performance of Lithium Germanium Phosphorus Sulfide (LGPS) SSEs [49].

By training the model with a small amount of data material, the computational scale and time scale of the model can be significantly increased compared to the original method, while the amount of calculation is reduced. Studies based on experimental data have certain advantages in terms of accuracy, but these advantages are limited to crystal structures that have been extensively researched. For most crystal structures, there are fewer known experimentally stable systems.

Therefore, in the current research environment, the approach of ML-assisted computation is undoubtedly superior to research based solely on experimental data.

2.2. Predicting the properties of SSEs

ML can analyze a vast amount of experimental data, identifying key factors that affect the performance of SSEs and aiding researchers in accelerating the prediction of properties of SSEs, thereby improving the accuracy of these predictions. High ionic conductivity in SSEs, one of the most critical indicators for assessing these materials, undoubtedly represents a major direction of current research.

For instance, Sendek et al. screened 12,831 lithium-containing crystalline solids to identify electrolytes with high structural and chemical stability, low electronic conductivity, and low cost. They then used logistic regression to develop a data-driven ionic conductivity classification model, predicting 21 materials that could exhibit rapid ionic conduction based on experimentally measured results reported in the literature [50]. Guo et al. combined DFT, ANN potentials, GA sampling, and ab initio molecular dynamics simulations to create a phase diagram for lithium thiophosphate (LPS) [51] (Fig. 4A). Based on trends discovered in the LPS phase diagram, they predicted SSEs with high ionic conductivity of $> 10^{-2} \text{ S cm}^{-1}$. Zhang et al. [52] used data from the Inorganic Crystal Structure Database (ICSD) and processed 528 modified X-ray diffraction (mXRD) quantitative representations as inputs to train their model. They confirmed that the mXRD encoding of anionic lattice symmetry and order in SSEs is closely related to ionic conductivity, predicting 16 new high lithium-ion conductivity SSEs (Fig. 4B). Tao et al. [53] used an unsupervised hierarchical clustering method to quickly predict 49 promising solid composite electrolytes with high ionic conductivity and discovered two structures with the lowest Li migration activation energy. Gabriel et al. [54] developed a method to predict the ionic conductivity of polymeric SSEs based on the electrolyte molecular structure and composition. They collected SSE data sets of polymers with known maximum ionic conductivity values from 217 experimental publications and used them for training. More than 20,000 potential polymer SSEs were screened using neural networks and XGBoost algorithms, the ionic conductivity values of thousands of candidate polymer SSEs were predicted, and further synthesis and characterization were performed.

The stability of SSE interfaces is also considered a significant predictive criterion. The exploration of their composition, structure, and function remains an unsolved challenge, particularly the issue of unintentional dendrite growth at interfaces, which can cause mechanical stress in SSEs. Thus, research into the interfaces and mechanical properties of SSEs is thriving. In 2018, Ahmad et al. [55] computationally screened over 12,000 inorganic solids, using a Graph Neural Network (GNN) model to predict the shear and bulk moduli of SSE materials (Fig. 4C). They also used Adaptive Boosting (AdaBoost), Least Absolute Shrinkage and Selection Operator (LASSO), and Bayesian ridge regression to predict the elastic constants of cubic crystal compounds, identifying more than 20 candidates for inhibiting dendrite growth in lithium metal and four types of SSEs based on mechanical anisotropy at the interface.

Jalem et al. [44] developed a single hidden layer feed-forward network with two external attributes, training it on 72 samples. In comparison with PLS regression, their feed-forward network model showed better predictive performance for investigating the diffusion barriers of SSEs. Cubuk et al. [56] used SVM for transfer learning, identifying 20 billion lithium-containing candidates, some of which could be potential SSEs.

ML plays a crucial role in screening and predicting SSEs, and with further increases in data and continuous optimization of algorithms, it is expected to reveal more about the properties and reaction mechanisms of SSEs.

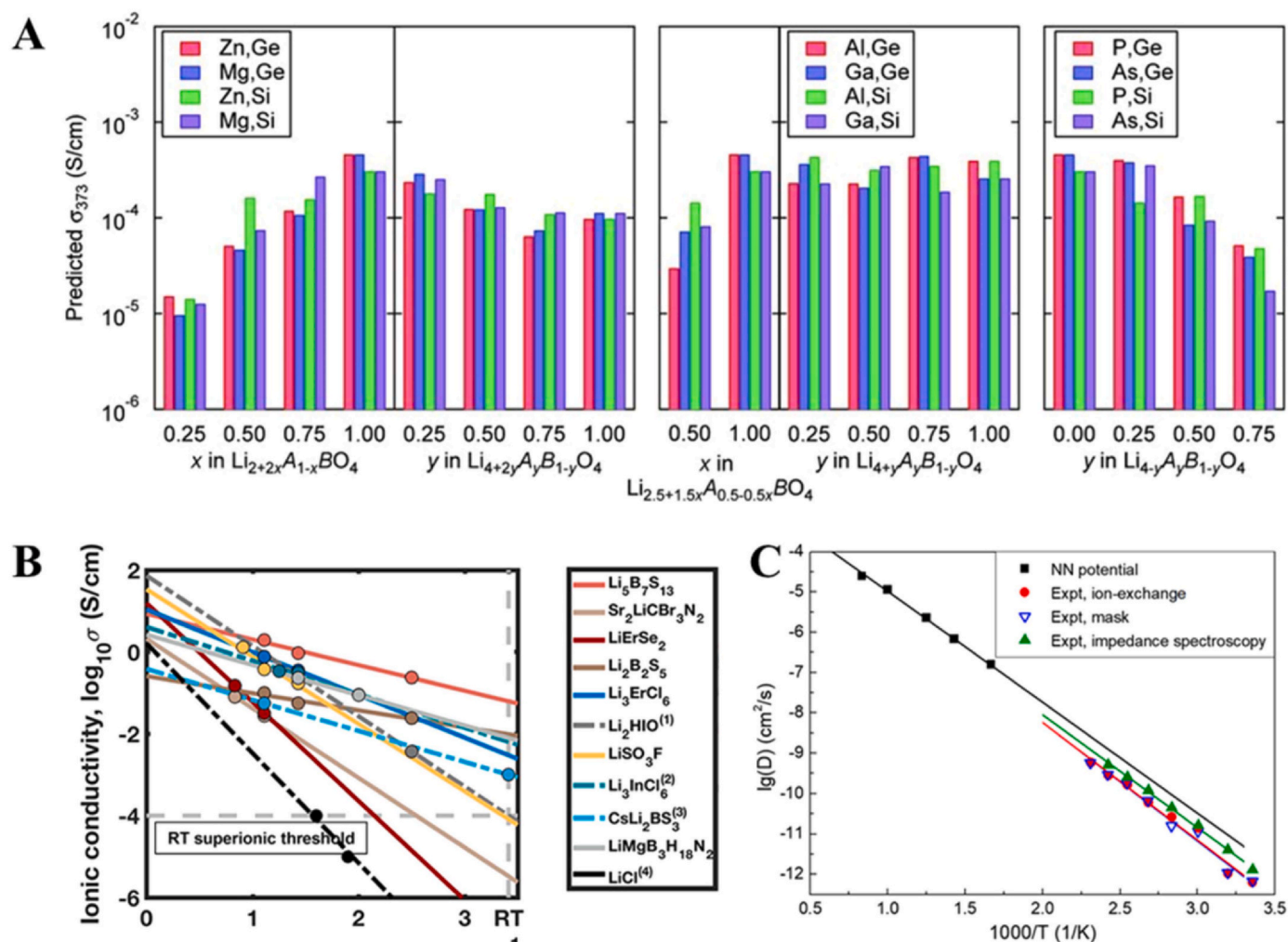


Fig. 3. (A) Ionic conductivity of 72 predicted components at 373 K. [41], Copyright 2013, Wiley Online Library. (B) Computationally observed ionic conductivity in candidate materials and extrapolation to room temperature. [42], Copyright 2019, American Chemical Society. (C) The diffusivity of Li in amorphous Li_3PO_4 . The simulation results were obtained from the large-scale MD using NN potentials. The experimental results were measured with different methods. [47], Copyright 2017, AIP Publishing.

(a) Reproduced with the permission from Ref. (b) Reproduced with the permission from Ref. (c) Reproduced with the permission from Ref.

2.3. Guiding the Synthesis of New SSEs

Chemical reactions involve interconnected data of products and intermediates. ML can also process structured data of chemical reactions, providing optimized strategies for synthesizing new SSEs and determining superior synthesis pathways to enhance the performance of SSEs. Ibrahim and Johan [57] studied the impact of chemical composition and temperature on the ionic conductivity of polymer electrolyte systems. They used Bayesian Neural Networks (BNN) to predict the ionic conductivity of polymer electrolyte systems, revealing how the ionic conductivity of these systems varies with chemical composition and temperature, and providing optimal solutions for improving the ionic conductivity of SSEs through synthesis temperature and chemical composition. A novel composite solid polymer electrolyte was successfully synthesized using a solution casting technique.

In comparison, the application of ML in guiding the synthesis of new SSEs has been much less, possibly due to the fact that most relevant information is proprietary to laboratories and not widely recorded. The lack of open databases related to SSE synthesis paths results in limited available data.

2.4. Optimizing Experimental Parameters

In traditional materials research, determining experimental parameters requires extensive trial and analysis. ML aids in rapidly predicting and forming iterative optimization design methods, optimizing experimental parameters, and guiding the acquisition of superior SSEs.

Beal et al. [58] used a high-throughput physical vapor deposition system to synthesize a library of 35 samples and studied perovskite-type SSEs. Within a broad parameter space (composition, thickness, deposition temperature, annealing temperature), an artificial neural network was constructed to determine the system's ionic conductivity behavior. It revealed that composition, especially lithium content and annealing temperature, are crucial factors in regulating the growth process (Fig. 5A). Chen et al. [59] employed principal component analysis (PCA), K-means clustering, and support vector machines (SVM) to analyze the relationship between manufacturing parameters and thin film quality. They emphasized the necessity of considering uniformity and ionic conductivity in the manufacturing process of SSE thin films, revealing the impact of manufacturing parameters on the quality of SSE

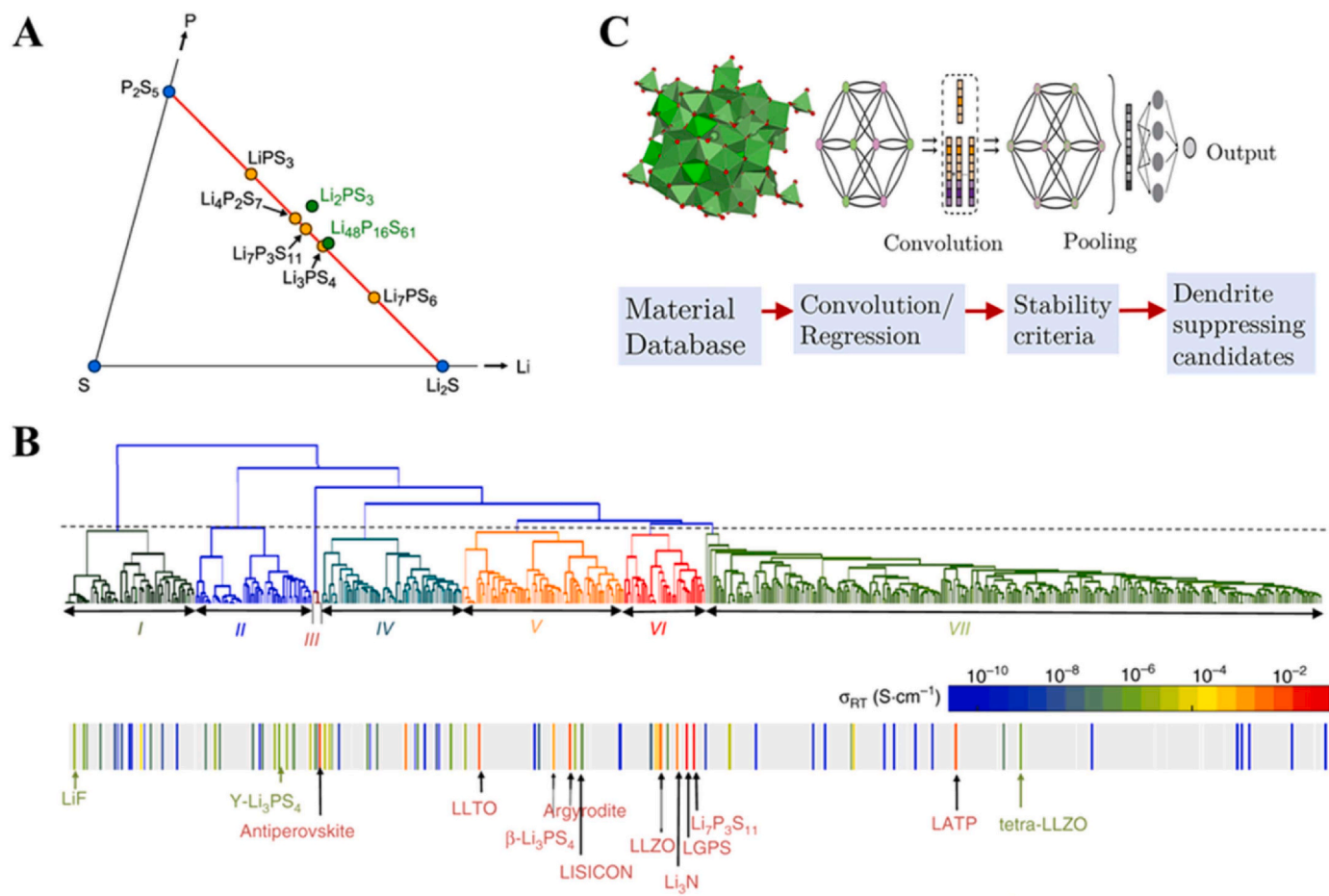


Fig. 4. (A) Excerpt from the ternary Li–P–S phase diagram showing reported LPS compositions on and near the Li_2S – P_2S_5 composition line. The materials falling on the right of the red line are sulfur-deficient compositions (green circles). [51], Copyright 2022, American Chemical Society. (B) Dendrogram generated by unsupervised clustering of all lithium-containing compounds and mapped to conductivity, divided into 7 groups. [52], Copyright 2019, Springer Nature. (C) Flowchart of a graph convolutional neural network. [55], Copyright 2018, American Chemical Society.

(a) Reproduced with the permission from Ref. (b) Reproduced with the permission from Ref. (c) Reproduced with the permission from Ref.

thin films. Luo et al. [60] used orthogonal experiments and ML methods to determine the sequence of important factors affecting the total ionic conductivity of LATP particles. They obtained the relationship between total ionic conductivity and sintering temperature, calcination source, and calcination temperature, identifying that sintering temperature is the most significant factor affecting ionic conductivity in the preparation process of this SSE (Fig. 5B).

2.5. Upgrading Characterization Methods

Utilizing ML to process and analyze complex characterization data can more accurately parse the microstructure and performance of materials.

Janis et al. [61] conducted statistical analysis based on ML-assisted sintering pellet image segmentation to quantify the microstructural

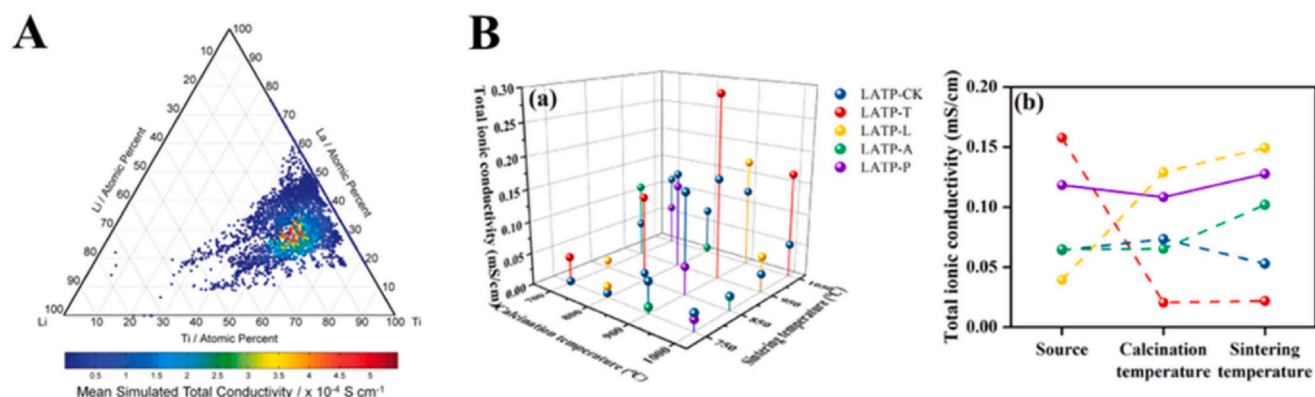


Fig. 5. (A) Predicted total conductivity as a function of composition. [58], Copyright 2011, American Chemical Society. (B) Total ionic conductivity of the LATP pellets and intuitive analysis of the orthogonal experiment. [60], Copyright 2023, American Chemical Society.

(a) Reproduced with the permission from Ref. (b) Reproduced with the permission from Ref.

changes in SSE grains and pore size with sintering time. Yang et al. [62] used unsupervised K-means clustering method analysis on hyperspectral Raman spectra, effectively revealing the mechanical in-situ cross-linking of polyethylene oxide and woven glass microscopy, a composite solid-state electrolyte with excellent performance. The strength results from the interaction of dynamic hydrogen and ionic bonds.

In contrast, data generated from characterization is multi-dimensional and complex, posing difficulties for standardization and ML training. Here, we only showcase ML applications in upgrading characterization methods applied to SSEs. In fact, ML have already significantly enhanced characterization methods for other materials, which we will not elaborate on further here.

In these various applications, improvements in one area can lead to improvements in another, creating an iterative cycle. Optimized computational simulation methods can enhance the ability to predict the properties of new solid-state electrolyte materials, while more effective experimental design and advanced new material synthesis characterization techniques can be fed back into the simulation and prediction process to continuously improve the entire research process. Especially important in this iterative loop [63]. The laboratory prepares solid electrolyte samples through chemical synthesis methods and then tests their physical properties and electrochemical performance. These experimental results are compared and verified with theoretical calculation predictions. If experimental results match computational predictions, this will increase confidence in computational methods and models. However, if the experimental results do not match the computational predictions, the computational model, parameter selection, or experimental conditions need to be re-examined to find out the reasons for the differences and adjust. During this process, experimental results and theoretical calculations feedback to each other, providing more accurate initial parameters and boundary conditions for the model, while also providing directions for improvement and adjustment for theoretical calculations. Through iterations of theoretical calculations and experimental verification, the theoretical model of solid electrolytes is gradually improved, and the accuracy and credibility of predictions are continuously improved. This systematic research approach, combining experiment and theory, jointly promotes the development of solid electrolyte materials [64,65].

Here, we have retrieved articles on the application of ML in SSEs and marked the relative popularity of each category of applications based on the number of articles (Table 1). We can see that currently, there is a relatively higher number of studies focused on optimizing simulation methods and screening and predicting properties related to SSEs. In contrast, research exploring the application of ML to enhance

the characterization methods of SSEs is relatively less. This observation provides a good research approach and direction for subsequent researchers in the field.

3. How to build database for ML

A prerequisite for the application of data-driven ML is the need for sufficient, accurate, and high-quality data to train the ML models. The amount of data available for different ML methods depends on the specific problem, and using high-quality data can prevent the consideration of incorrect or redundant information, which will directly impact the performance of the ML models [66].

The main types of data applied in ML for SSEs broadly include data from computational simulations, experimental material property data, literature data, and image data.

Over the past few decades, databases in materials science have evolved towards digitization. Publicly open-source databases have cataloged the crystal structures of already synthesized compounds, and computational material data have been categorized into various databases. Common public databases used in SSE research include the Inorganic Crystal Structure Database (ICSD) [67], Crystallography Open Database (COD) [68], Cambridge Structural Database (CSD) [69], Materials Project [70], Open Quantum Materials Database (OQMD) [71], Novel Materials Discovery (NOMAD) [72], and Materials Cloud [73], among others. These databases have been mentioned in many excellent reviews [74,75] In addition to these well-known and widely used material databases, recent years have seen the emergence of databases like Matgen [76], which contains crystal structure information and ion migration channel connectivity. IonML [77], a ML platform for designing superionic conductors. The Liverpool Ionics Dataset, evaluation of experimentally measured lithium SSEs conductivity database using ML [78]. And databases for simulating the sulfur K-edge X-ray absorption spectra of lithium thiophosphate SSEs [79]. The continuous appearance of new SSE-related databases indicates the growing interest in the field of SSEs.

Researchers face challenges in effectively managing, cleaning, and utilizing the data from these vast databases. Despite containing rich information, scientific databases may mislead subsequent research due to human errors and measurement inaccuracies during the recording process. Additionally, since databases only record a small portion of all possible chemical structures, more intelligent methods are needed to expand and optimize the use of data. Management and cleaning of experimental data are important tasks in scientific research [80]. The quality of data directly affects the performance of ML models, as their

Table 1
Current popularity of relevant literature for five types of applications.

	Popularity	Electrolyte System	Resolve	Algorithms	Example (Ref.)
Optimizing Computational Simulation Methods		LISICON	Optimizing the computational process, reduces the amount of computational.	Logistic regression	[42]
Predicting the properties of SSEs		LISICON	Predicting the lithium diffusion barrier.	PLS, Neural Network	[44]
Guiding the Synthesis of New SSEs		Nanocomposite polymer	Explore the effects of chemical composition and temperature on performance and perform synthesis.	BNN	[57]
Optimizing experimental parameters		Thin Film SSE	Optimization of manufacturing parameters related to quality.	K-means, SVM, PCA	[59]
Upgrading characterization methods		LLZO	Aided SEM image segmentation for statistical analysis and quantification of microstructures in sintering processes	-	[61]

Popularity increases as color deepens.

training and predictive capabilities highly depend on accurate and reliable data inputs. The statistical properties of ML can be used to identify and correct anomalies in the data, improving its accuracy. Data cleaning not only helps researchers avoid misdirection but also enhances research efficiency and prevents the wastage of resources [81].

The completeness of experimental data plays a crucial role in scientific research, particularly in materials science. Every experimental step and each observation result, whether successful or not, may contain key information that drives scientific progress. However, in traditional experimental science practices, a significant portion of data is often not fully recorded, especially for experiments that do not achieve the expected results, or “failures”. This approach overlooks a critical fact: failed experiments can also provide valuable insights for future research [82]. For instance, a “failed” synthesis experiment might reveal the instability of a certain material under specific conditions, information that is vital for adjusting the synthesis pathway. Unfortunately, researchers often exhibit bias in data recording, tending to document only those results that align with expectations or are deemed “successful”. This selective recording method not only overlooks the potential value of failed experiments but can also lead to a significant waste of resources. When other research teams repeat the same experimental paths, the lack of information about previous failures may lead them to make the same mistakes. Therefore, establishing a system that comprehensively records experimental data is particularly important. In this regard, open-source Electronic Laboratory Notebooks (ELNs) provide a solution [83]. Such platforms not only allow researchers to collaboratively record experimental data but also integrate with laboratory instruments for real-time data collection and automatic recording. This way, even results that do not meet expectations are fully documented, providing reference for future research. Fig. 6 shows the architecture of ELN.

The integration of literature data and image data still presents challenges. These data are often stored disparately and in varying formats. Without suitable computational tools, researchers must spend considerable time manually filtering and integrating this data, a process that is not only inefficient but also prone to errors. To overcome this challenge, text mining technology has emerged, enabling automatic extraction of relevant data and information from a large volume of literature. Text mining can use methods such as Natural Language Processing (NLP) to understand and interpret the language in academic literature, thus efficiently extracting data [84]. This technology can be applied not only to textual information but also to the analysis of image data. Through advanced image recognition and processing techniques, ML models can extract valuable information from image data, such as structural features of materials and changes in phase diagrams [85].

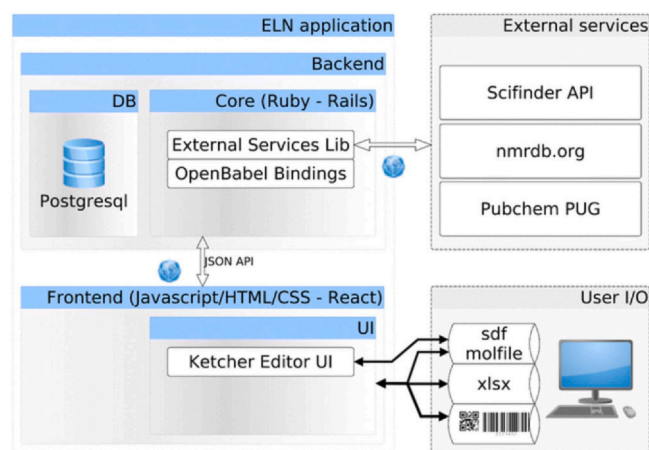


Fig. 6. ELN architecture diagram: summary of programming languages, front-end input/output, and external service connection. Reproduced with the permission from Ref. [83], Copyright 2017, Springer Nature.

The database for SSEs applied in ML can be constructed using the methods. However, it is evident that these data are stored in various databases, each with its diverse and scattered format, making the handling of data from multiple sources quite complex. Against this backdrop, the FAIR principles for data management have been proposed, ensuring that data are Findable, Accessible, Interoperable, and Reusable. This means that the data should not only be easy to discover and access, but also stored in a standardized and universal format, facilitating effective utilization by various researchers and algorithms. Implementing these principles will maximize the added value of the data, benefiting the entire academic community [86].

Building an SSE-related database needs to contain enough material property information, such as chemical composition, crystal structure, lattice constants, kinetic and thermodynamic properties, etc., to comprehensively evaluate the material properties of SSEs. The database should ensure reliable sources, and theoretical calculation results should be verified and complemented with experimental data to provide a comprehensive performance evaluation. Documentation of experimental methods and conditions for each performance is necessary to ensure comparability of data. In model construction, when applying ML to the SSE field through different algorithm models, the SSE data used will also be different. Most ML models are nonlinear, so model output results are optimized by improving the model or employing different feature engineering until a higher performance index is achieved. Therefore, an important feature of the SSE database is the need to contain comprehensive and accurate original data, such as databases such as Matgen [76], to ensure that the results when ML is applied to SSE are accurate enough to provide an accurate data basis for subsequent model construction.

4. Model building

4.1. Featurization

After acquiring a large amount of data related to SSEs, the presentation form of these data has a significant impact on the ML process. In this process, feature engineering becomes a key step, involving the extraction of key features from the original database and applying these features to ML algorithms. The material-related data we extract need to be encoded and transformed into a format that computers can understand, namely “descriptors”. This practice of mapping material property data to descriptors suitable for the ML process is crucial for the application of ML [87].

When selecting descriptors, it is necessary to consider their compatibility with the ML model used. Different ML algorithms may have significant differences in data requirements and processing methods. For example, some algorithms may be better at handling high-dimensional data, while others may be more suitable for low-dimensional or sparse data. Therefore, choosing descriptors that match a specific algorithm helps to improve the model's efficiency and accuracy. Moreover, the diversity of descriptors is a factor that cannot be ignored. When dealing with SSE materials, since their complex structures and properties are affected by multiple factors, relying on a single type of descriptor may not fully capture the multi-faceted properties of the material. Therefore, multiple descriptors need to be comprehensively utilized to comprehensively evaluate the performance of SSE materials. The combined use of various types of descriptors often provides more comprehensive information, thereby improving the accuracy and robustness of the predictive model [88]. There are many excellent reviews analyzing the descriptors used in ML for SSEs [89–93], and this field remains an active area of research.

The selection of descriptors, whether driven by domain knowledge or data-driven, aims to build more effective and interpretable models. Domain knowledge-driven descriptors, such as electronegativity, atomic number, ionic radius, lattice constants, cell size, etc., rely on fundamental concepts and theories of material science. Data-driven

Table 2

ML Methods Applied to SSEs and Advantages and Disadvantages [92,102–105].

Model Abbreviation	Model Full Name	Advantages	Disadvantages
LR	Linear Regression	Simple and interpretable. Performs well on linear relationship problems.	Poor fit for non-linear relationships. Sensitive to outliers.
Lasso Regression	Lasso Regression	Handles high-dimensional data. Controls model complexity.	May lead to sparse coefficients, challenging interpretation.
EN	Elastic Net	Manages multicollinearity. Suitable for feature selection.	Sensitive to outliers. Requires tuning regularization parameters.
GPR	Gaussian Process Regression	Applicable to small sample sizes. Non-linear modeling capability.	High computational complexity. Challenging for large-scale data.
NB	Naive Bayes	Simple and fast. Suitable for high-dimensional data.	Assumes feature independence.
SVM	Support Vector Machine	Suitable for high-dimensional data. Performs well on complex classification tasks. Effective for small sample sizes.	High computational cost. Kernel choice requires expertise.
DT	Decision Tree	Easy to understand and visualize. Handles non-linear relationships.	Prone to overfitting. Sensitive to minor data variations.
RF	Random Forest	Applicable to high-dimensional data. Good interpretability.	Longer training and prediction times. Requires many trees for optimal performance.
XGBoost	Extreme Gradient Boosting	Efficient training and prediction. Good interpretability.	Requires parameter tuning. Needs more data.
SVR	Support Vector Regression	Effective in high dimensional spaces. Robust against overfitting in high-dimensional space	Computationally intensive. Sensitive to the choice of kernel and regularization parameters
AdaBoost	Adaptive Boosting	Improves the accuracy of weak classifiers. Simple to implement. Less prone to overfitting	Sensitive to noisy data and outliers. Can be slow to train
ET	Extra Trees	Randomizes choices, reducing overfitting. Faster training than Random	May perform worse than Random Forests on noisy tasks.
KNN	K-Nearest Neighbors	Simple and intuitive. Effective for multi-class problems.	Computationally expensive with large datasets. Sensitive to imbalanced data and missing values.
LGBM	Light Gradient Boosting Machine	Fast training efficiency. Good performance on large datasets.	More prone to overfitting compared to some other models
GB	Gradient Boosting	Powerful predictive ability.H	More prone to overfitting than Random Forests
PLS	Partial Least Squares	Suitable for high-dimensional data. Reduces data dimensionality.	Sensitive to outliers. Model complexity.
WVR	Weighted Voting Regression	Combines predictions from multiple models. Can handle heterogeneous data.	Requires selecting appropriate base models. Complexity.
PCA	Principal Component Analysis	Reduces data dimensionality, extracts main features. Visualizes data.	Sensitive to outliers.
K-means	K-means Clustering	Simple and easy to understand. Efficient for large-scale data.	Requires choosing the number of clusters. Sensitive to initial centroids.
GNN	Graph Neural Networks	Suitable for Graph Data. Scalability. Adaptability.	High computational complexity. Prone to overfitting. Challenges with large-scale graphs.
FFNN	Feedforward Neural Network	Easy to train and implement. Applicable to various deep learning tasks.	May require many hidden layers and neurons for some tasks. Potential for overfitting.
RBFNN	Radial Basis Function Neural Network	Demonstrates good generalization on some problems. Can be used for function approximation.	Complex model parameter selection. Requires a large amount of data for parameter tuning.
BNN	Bayesian Neural Network	Provides uncertainty estimates. Helps prevent overfitting.	Sensitive to choose of RBF functions. High computational complexity. May be unstable with small datasets.

descriptors, on the other hand, use algorithms to extract information from a large amount of data to form a comprehensive description of the crystal structure. This includes the Coulomb matrix [94], Atomic Centered Symmetry Functions (ACSF) [95], Smooth Overlap of Atomic Positions (SOAP) [96], etc.

However, the feature selection process driven by domain knowledge may have the possibility of subjective bias, which can lead to blind spots in the model. For example, certain important physical or chemical properties may be overlooked because they are not sufficiently emphasized in traditional theories. In contrast, data-driven approaches are more objective and comprehensive in feature selection, but they also pose challenges. For instance, the quality and quantity of data have a significant impact on the performance of the model. Inaccurate or incomplete data may lead the model to learn incorrect or misleading patterns.

Currently, in the field of ML applied to SSEs, there is a trend to follow the application in other materials, choosing to generate their own feature sets from a large amount of unstructured initial feature sets through deep learning methods. This eliminates the manual feature engineering and representation steps in traditional ML. Although deep

learning has achieved tremendous success in the applications, its application in the field of material science is still in its early stages, mainly limited by the scarcity of large training datasets. Nevertheless, deep learning has already shown certain potential in SSEs. For example, Crystal Graph Convolutional Neural Networks (CGCNN) have been used to predict the mechanical properties of SSEs [97]. Given the current success of deep learning in machine vision and language translation, it is expected that feature maps based on deep learning will play an important role in future ML applications.

4.2. The Choice of Model Approach is Crucial

With data in the appropriate format, we can meet our needs by building ML models. In fact, the materials science community has just begun to explore the application of ML, where the extent of algorithm usage is largely driven by specific requirements. The choice of model depends on the nature of the research question and the characteristics of the model itself. In materials science applications, many have introduced suitable ML algorithms [98–101]. For example, when predicting electrical conductivity properties, linear regression may be

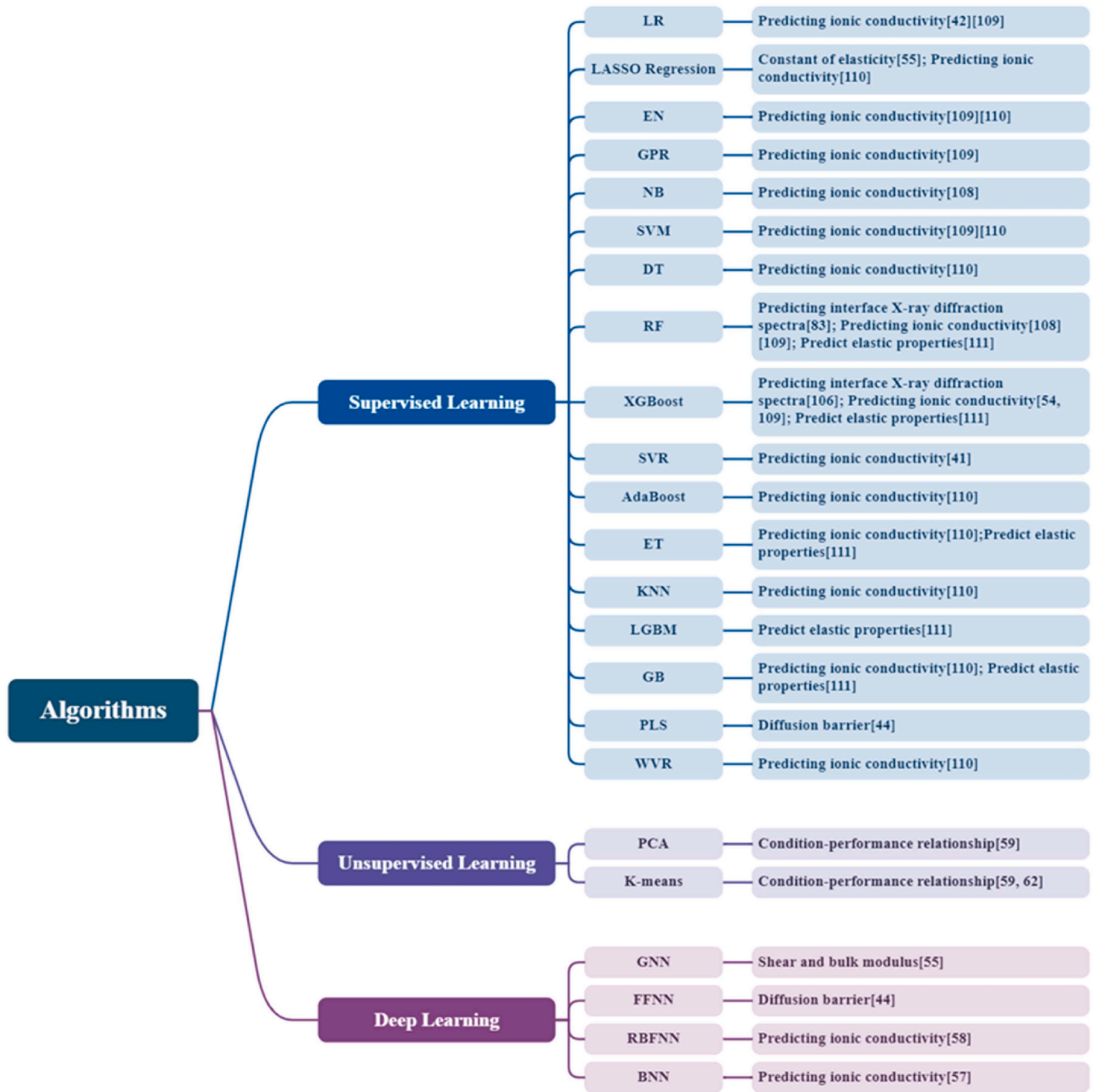


Fig. 7. A mind map of ML methods applied to various aspects of SSEs.

chosen; for classification tasks, such as the thermodynamic stability of materials, logistic regression or support vector machines may be required. Simple linear models may be sufficient for most materials, but SSEs with complex structures may also require more sophisticated models, such as deep learning. Here, we review ML success stories in the field of SSE. Specifically, we will discuss the process of training ML models on constructed datasets, how to screen and select the best models, and how to validate the results of these models. Here, we review successful cases of ML in the field of SSEs. Specifically, we will discuss the process of training ML models on constructed datasets, how to screen and select the best models, and how to verify the results of these models.

Supervised learning, unsupervised learning, and reinforcement learning are the three main types in the field of ML. Supervised learning

is the most common type of ML model in SSE research. In this type of learning, the model predicts the output of new data by learning from labeled training data. For example, the model might learn how to predict attributes such as the mechanical properties of SSEs from their structural composition [50]. Unsupervised learning does not rely on labeled data, but rather explores the structure of the data itself. In SSE research, unsupervised learning can be used to identify patterns and relationships in data, such as using cluster analysis to identify associations between material properties [60]. Reinforcement learning is a relatively new field and has not yet been widely applied in SSE research. It mimics human learning by learning optimal strategies through interaction with the environment. In recent years, with the improvement of computing power, deep learning methods have been

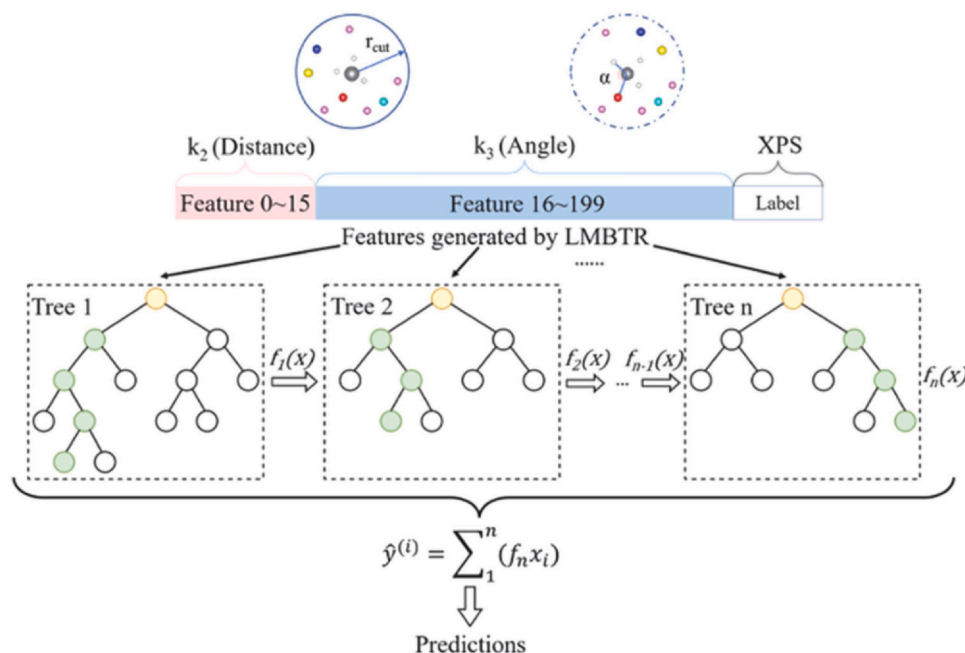


Fig. 8. The model of XGBoost used in XPS prediction. Reproduced with the permission from Ref. [83], Copyright 2022, American Chemical Society.

used to handle more complex pattern recognition tasks built on supervised learning. In the study of SSEs, deep learning can be used to extract meaningful information from complex datasets, such as analyzing the microstructure of materials using image recognition techniques [55].

In the previous text, we exemplified the specific applications of various ML algorithms in SSEs. Here, we have organized the classification of ML algorithms applied in SSEs, marked the advantages and disadvantages of each ML algorithm (Table 2), and we made a guided map of ML methods applied to various aspects of SSEs (Fig. 7).

In the early stages of ML applications in SSEs, selecting and testing a variety of different models is a common and effective practice. Sun et al. used four models: linear regression, random forest (RF), neural networks (NN), and extreme gradient boosting (XGBoost), for predicting the X-ray photoelectron spectra at SSE interfaces [106], as shown in Fig. 8 (best model only). Similarly, Kim et al. proposed a ML-based ensemble model specifically for the preliminary screening of the thermodynamic stability of sodium superionic conductors (NASICON materials) [107]. Kang et al. employed nine different ML models, including naive Bayes (NB), logistic regression, decision trees (DT), K-nearest neighbors (KNN), RF, gradient boosting (GB), stochastic gradient descent (SGD), Light Gradient Boosting Machine (LGBM), and XGBoost, to screen SSEs for lithium-ion batteries [108]. Mandal et al. utilized linear regression, elastic net (EN), Lasso regression, support vector machines (SVM), XGBoost, NN, RF, and Gaussian process regression (GPR) to study the impact of different glass compositions on the ionic conductivity of sodium ion conducting glass electrolytes [109]. Xu et al. used a weighted voting regression (WVR) model to predict the ionic conductivity of oxides, involving models such as AdaBoost, DT, EN, Extra Trees (ET), GB, KNN, Lasso regression, RF, and SVM [110]. Sun et al. developed a ML-based model to predict the elastic properties of $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ derived materials, trained with five models including RF, ET, GB, XGBoost, and LGBM [111]. Each model has its unique strengths and limitations, and choosing the appropriate model is crucial to achieving research objectives.

Once the application of ML methods has been established, the next steps are the training and evaluation of the model. Many technical details for training and validating models are well introduced in several reviews [112–114]. The main differences in SSE compared to other materials are not significant, and here we will not review further

the subsequent evaluation of ML models applied to SSEs. However, it is worth emphasizing that the application of ML models in the field of SSEs has unique challenges and considerations, which can have a significant impact on model optimization and final results. For example, data related to SSEs often have high-dimensional features, and special attention needs to be paid to avoid overfitting and ensure the generalizability of the model. When using ML to explore SSEs, one often faces the conflict of high dimensions with small data size [115]. In the face of this dilemma, it is necessary to consider both data and algorithms. Before computing, more data should be mined, and when selecting and designing algorithms, the characteristics of the dataset need to be considered. For high-dimensional small-size datasets, simple yet effective algorithms (such as linear regression, logistic regression, SVM, etc.) are often more appropriate. Additionally, active learning and transfer learning can be chosen for processing. In the case of limited datasets, active learning can effectively select or generate the most informative data samples for labeling and training. This method can significantly improve learning efficiency by prioritizing the learning of data that can most improve model performance. Transfer learning is another powerful tool for handling small datasets. By transferring models already trained on related tasks to SSE datasets, one can utilize features and patterns already learned by the pre-trained models to achieve better performance on limited data [116,117]. Fig. 9 shows the workflow of active and transfer learning in materials science.

During the model validation process, it is particularly important to ensure that the model performs well not only on the training data but also maintains stable performance on new, unseen data. This requires comprehensive model evaluation, including testing with independent test datasets and using cross-validation methods to assess the generalizability of the model. Additionally, the interpretability of the model is very crucial, especially in the field of materials science, where the interpretability of the model helps to deepen the understanding of the properties and behaviors of materials.

In summary, the application of ML in SSE research is a field full of challenges but also full of opportunities. By properly handling data, selecting appropriate models, and conducting thorough validation and testing, ML can not only accelerate the discovery and optimization of electrolyte materials but also provide new avenues for a deeper understanding of these materials.

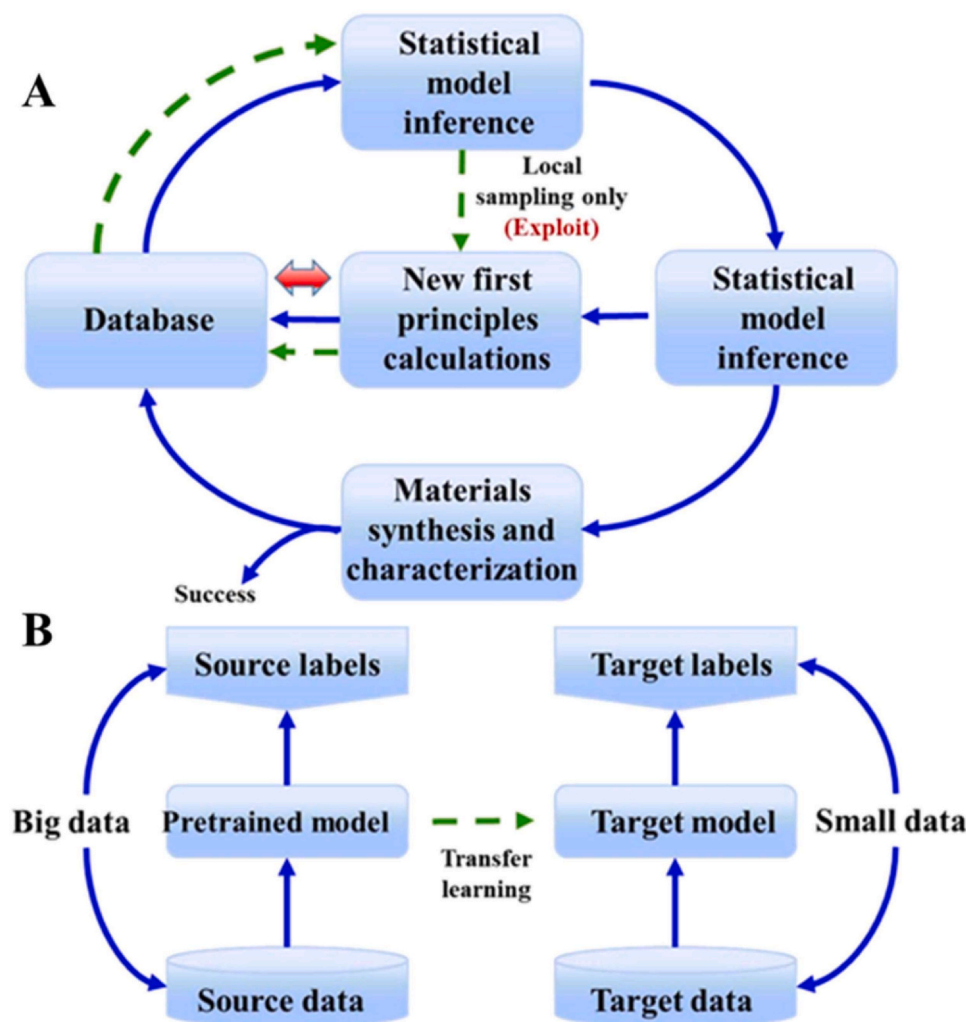


Fig. 9. The workflows of (A) active learning and (B) transfer learning in materials science. Reproduced with the permission from Ref. [116], Copyright 2023, Springer Nature.

5. Conclusion and outlook

Reflecting on the application and development of ML in the field of SSEs, we recognize that although ML has made significant progress in modeling and prediction, it still faces some limitations in practical application and validation. Although various successful models have been proposed in the literature, mainly focusing on data-driven analysis and pattern recognition, the decisive role of ML in the actual synthesis and development of SSEs has not yet been fully realized.

As data-intensive science (the fourth paradigm of science) continues to evolve, we are entering the fifth paradigm, centered on knowledge and emphasizing intelligence-driven science [117]. Under this new paradigm, ML is expected to integrate more deeply into the entire research and development process of SSEs. We anticipate that ML will play a more direct role in the experimental and application phases of SSEs. The knowledge-centered, intelligence-driven platform will directly control the experimental process through ML and achieve real-time attribute prediction. Advanced ML models will continuously monitor multiple parameters during SSE synthesis, including chemical reactions and voltages, as well as the interactions between these parameters. These models will be able to optimize multiple attributes simultaneously, using multi-objective optimization methods to consider trade-offs between attributes to achieve optimal results [118]. In addition, the use of multi-scale modeling technology can clarify the transmission characteristics and complex interface relationships

between SSE and batteries, provide a macro perspective for micro-scale systems, and comprehensively optimize SSE. Through integration with laboratory automation, robots can conduct experiments and make precise measurements, enabling high-throughput experiments. This automated experiment system will transform the experimental process from traditional slow-paced manual experiments to automated and efficient experiments [119]. In this way, the synthesis and prediction of solid electrolytes will form an iterative cycle that continuously accelerates the entire research process. Such an approach will help develop more efficient and longer-lasting energy storage solutions.

The rise of the ChatGPT large language model [120] has created a global sensation. Currently, the design and synthesis of new materials without human assistance, using ML and experimental robots, has been achieved, with deep learning significantly accelerating the discovery of new materials. By combining ML algorithms, experimental robots can not only automatically execute experimental procedures but also self-adjust and optimize based on experimental results, greatly enhancing experimental efficiency and precision. In an automated, intelligent laboratory, ML and experimental robots collaborate to continuously explore and discover new materials [121].

To fully unleash the potential of ML in the field of SSEs, more interdisciplinary collaboration is needed. This includes experts in materials science, computer science, chemistry, and physics working together to further accelerate the design of potential solutions for SSEs and promote the joint development of SSEs and ML. Interdisciplinary

teams can collectively tackle challenges in data collection, model development, and experimental validation, making the application of ML methods in SSE research more efficient and accurate.

In the future, with the emergence of more advanced ML models and algorithms, combined with big data analytics and automated experimental techniques, we have reason to believe that ML will play an even more critical role in future SSE research. This will not only accelerate the discovery of new electrolyte materials but also drive innovation in battery technology, offering more efficient and safer solutions for energy storage and conversion.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The work was financially supported by the National Natural Science Foundation of China (12175089, 12205127, 52203303, 61976156, 52220105010), the Key Research and Development Program of Yunnan Province (202103AF140006), the Applied Basic Research Programs of Yunnan Provincial Science, the Technology Department (202001AW070004, 202301AS070051, 202301AU070064), the Yunnan Major Scientific and Technological Projects (202202AG050003), the Natural Science Foundation of Guangdong Province (2022A1515010076), the Shenzhen Science and Technology Innovation Program (SGDX20211123151002003), the German Research Foundation (DFG: LE 2249/15-1) and the Sino-German Centre for Research Promotion (GZ1579).

References

- [1] Y. Pang, J. Pan, J. Yang, S. Zheng, C. Wang, Electrolyte/electrode interfaces in all-solid-state lithium batteries: a review, *Electrochem. Energy Rev.* 4 (2021) 169–193.
- [2] D. Larcher, J.-M. Tarascon, Towards greener and more sustainable batteries for electrical energy storage, *Nat. Chem.* 7 (2015) 19–29.
- [3] Y.Z. Zhu, X.F. He, Y.F. Mo, Origin of outstanding stability in the lithium solid electrolyte materials: insights from thermodynamic analyses based on first-principles calculations, *ACS Appl. Mater. Interfaces* 7 (2015) 23685–23693.
- [4] R. Gond, W. Van Ekeren, R. Mogensen, A.J. Naylor, R. Younesi, Non-flammable liquid electrolytes for safe batteries, *Mater. Horiz.* 8 (2021) 2913–2928.
- [5] K. Liu, Y. Liu, D. Lin, A. Pei, Y. Cui, Materials for lithium-ion battery safety, *Sci. Adv.* 4 (2018) eaas9820.
- [6] X. Chen, W. He, L.-X. Ding, S. Wang, H. Wang, Enhancing interfacial contact in all solid state batteries with a cathode-supported solid electrolyte membrane framework, *Energy Environ. Sci.* 12 (2019) 938–944.
- [7] X. Miao, S. Guan, C. Ma, L. Li, C.W. Nan, Role of interfaces in solid-state batteries, *Adv. Mater.* 35 (2023) 2206402.
- [8] L. Zhou, T. Zuo, C.Y. Kwok, S.Y. Kim, A. Assoud, Q. Zhang, J. Janek, L.F. Nazar, High areal capacity, long cycle life 4V ceramic all-solid-state Li-ion batteries enabled by chloride solid electrolytes, *Nat. Energy* 7 (2022) 83–93.
- [9] X. Judez, H. Zhang, C. Li, G.G. Eshetu, J.A. González-Marcos, M. Armand, L.M. Rodríguez-Martínez, Review-solid electrolytes for safe and high energy density lithium-sulfur batteries: promises and challenges, *J. Electrochem. Soc.* 165 (2018) A6008–A6016.
- [10] Y. Cai, C. Li, Z. Zhao, D. Mu, B. Wu, Air stability and interfacial compatibility of sulfide solid electrolytes for solid-state lithium batteries: advances and perspectives, *ChemElectroChem* 9 (2022) e202101479.
- [11] R. Chen, W. Qu, X. Guo, L. Li, F. Wu, The pursuit of solid-state electrolytes for lithium batteries: from comprehensive insight to emerging horizons, *Mater. Horiz.* 3 (2016) 487–516.
- [12] Y. Morino, S. Kanada, Degradation analysis by X-ray absorption spectroscopy for LiNbO₃ coating of sulfide-based all-solid-state battery cathode, *ACS Appl. Mater. Interfaces* (2023).
- [13] Z. Xue, D. He, X. Xie, Poly (ethylene oxide)-based electrolytes for lithium-ion batteries, *J. Mater. Chem. A* 3 (2015) 19218–19253.
- [14] G. Bucci, B. Talamini, A.R. Balakrishna, Y.M. Chiang, W.C. Carter, Mechanical instability of electrode-electrolyte interfaces in solid-state batteries, *Phys. Rev. Mater.* 2 (2018).
- [15] M.J. Lee, J. Han, K. Lee, Y.J. Lee, B.G. Kim, K. Jung, B.J. Kim, S.W. Lee, Elastomeric electrolytes for high-energy solid-state lithium batteries, *Nature* 601 (2022) 217–222.
- [16] J. Janek, W.G. Zeier, Challenges in speeding up solid-state battery development, *Nat. Energy* 8 (2023) 230–240.
- [17] Y. Lu, C. Zhao, J. Huang, Q. Zhang, The timescale identification decoupling complicated kinetic processes in lithium batteries, *Joule* 6 (2022) 1172–1198.
- [18] Y. Ma, J. Wan, Y. Yang, Y. Ye, X. Xiao, D.T. Boyle, W. Burke, Z. Huang, H. Chen, Y. Cui, Z. Yu, S.T. Oyakhire, Y. Cui, Scalable, ultrathin, and high-temperature-resistant solid polymer electrolytes for energy-dense lithium metal batteries, *Adv. Energy Mater.* 12 (2022).
- [19] Z. Ning, G. Li, D. Melvin, Y. Chen, J. Bu, D. Spencer-Jolly, J. Liu, B. Hu, X. Gao, J. Perera, C. Gong, S.D. Pu, S. Zhang, B. Liu, G.O. Hartley, A.J. Bodey, R.I. Todd, P.S. Grant, D. Armstrong, T.J. Marrow, C.W. Monroe, P.G. Bruce, Dendrite initiation and propagation in lithium metal solid-state batteries, *Nature* 618 (2023) 287–293.
- [20] B. Hu, J. Xu, Z. Fan, C. Xu, S. Han, J. Zhang, L. Ma, B. Ding, Z. Zhuang, Q. Kang, X. Zhang, Covalent organic framework based lithium-sulfur batteries: materials, interfaces, and solid-state electrolytes, *Adv. Energy Mater.* 13 (2023).
- [21] S. Curtarolo, G.L.W. Hart, M.B. Nardelli, N. Mingo, S. Sanvito, O. Levy, The high-throughput highway to computational materials design, *Nat. Mater.* 12 (2013) 191–201.
- [22] J. Wei, X. Chu, X.Y. Sun, K. Xu, H.X. Deng, J. Chen, Z. Wei, M. Lei, Machine learning in materials science, *InfoMat* 1 (2019) 338–358.
- [23] A. Choudhary Agrawal, Perspective: Materials informatics and big data: realization of the “fourth paradigm” of science in materials science, *APL Mater.* 4 (2016) 053208.
- [24] Q. Hu, K. Chen, F. Liu, M. Zhao, F. Liang, D. Xue, Smart materials prediction: applying machine learning to lithium solid-state electrolyte, *Materials* 15 (2022) 1157.
- [25] S.M. Moosavi, K.M. Jablonka, B. Smit, The role of machine learning in the understanding and design of materials, *J. Am. Chem. Soc.* 142 (2020) 20273–20287.
- [26] L. Xu, L. Wencong, P. Chunrong, S. Qiang, G. Jin, Two semi-empirical approaches for the prediction of oxide ionic conductivities in ABO₃ perovskites, *Comput. Mater. Sci.* 46 (2009) 860–868.
- [27] <https://www.mgi.gov/>.
- [28] K. Rajan, Materials informatics: the materials “Gene” and big data, *Annu. Rev. Mater. Res.* 45 (2015) 153–169.
- [29] J.-P. Correa-Baena, K. Hippalgaonkar, J. Van Duren, S. Jaffer, V.R. Chandrasekhar, V. Stevanovic, C. Wadia, S. Guha, T. Buonassisi, Accelerating materials development via automation, machine learning, and high-performance computing, *Joule* 2 (2018) 1410–1420.
- [30] L. Ward, M. Aykol, B. Blaiszik, I. Foster, B. Meredig, J. Saal, S. Suram, Strategies for accelerating the adoption of materials informatics, *MRS Bull.* 43 (2018) 683–689.
- [31] Y.L. Liu, C. Niu, Z. Wang, Y. Gan, Y. Zhu, S.H. Sun, T. Shen, Machine learning in materials genome initiative: a review, *J. Mater. Sci. Technol.* 57 (2020) 113–122.
- [32] K.T. Butler, D.W. Davies, H. Cartwright, O. Isayev, A. Walsh, Machine learning for molecular and materials science, *Nature* 559 (2018) 547–555.
- [33] S. Hu, C. Huang, Machine-learning approaches for the discovery of electrolyte materials for solid-state lithium batteries, *Batteries* 9 (2023) 228.
- [34] S. Pereznieto, R. Jaafreh, J. Kim, K. Hamad, Discovery of solid-state electrolytes for Na-ion batteries using machine learning, *Mater. Lett.* 349 (2023) 134848.
- [35] Y. Qiu, X. Zhang, Y. Tian, Z. Zhou, Machine learning promotes the development of all-solid-state batteries, *Chin. J. Struct. Chem.* 42 (2023) 100118.
- [36] T.T. Vu, H.J. Cheon, S.Y. Shin, G. Jeong, E. Wi, M. Chang, Hybrid electrolytes for solid-state lithium batteries: challenges, progress, and prospects, *Energy Storage Mater.* 61 (2023) 102876.
- [37] J. Tian, Y. Fan, T. Pan, X. Zhang, J. Yin, Q. Zhang, A critical review on inconsistency mechanism, evaluation methods and improvement measures for lithium-ion battery energy storage systems, *Renew. Sustain. Energy Rev.* (2024) 113978.
- [38] M.K. Tufail, P. Zhai, W. Khokar, M. Jia, N. Zhao, X. Guo, Evaluation of solid electrolytes: development of conventional and interdisciplinary approaches, *Interdiscip. Mater.* 2 (2023) 529–568.
- [39] C.B. Eng, Y. Soni, E. Lum, Z. Khoo, S.K. Yao, V. Vineeth, J. Kumar, C.S. Lu, C. Johnson, Z.W. Seh Wolverton, Theory-guided experimental design in battery materials research, *Sci. Adv.* 8 (2022) eabm2422.
- [40] G. Hautier, C. Fischer, V. Ehrlicher, A. Jain, G. Ceder, Data Mined ionic substitutions for the discovery of new compounds, *Inorg. Chem.* 50 (2011) 656–663.
- [41] K. Fujimura, A. Seko, Y. Koyama, A. Kuwabara, I. Kishida, K. Shitara, C.A.J. Fisher, H. Moriwake, I. Tanaka, Accelerated materials design of lithium superionic conductors based on first-principles calculations and machine learning algorithms, *Adv. Energy Mater.* 3 (2013) 980–985.
- [42] A.D. Sendek, E.D. Cubuk, E.R. Antoniak, G. Cheon, Y. Cui, E.J. Reed, Machine learning-assisted discovery of solid Li-ion conducting materials, *Chem. Mater.* 31 (2019) 342–352.
- [43] R. Jaleem, T. Aoyama, M. Nakayama, M. Nogami, Multivariate method-assisted ab initio study of olivine-type LiMxO₄ (main group M²⁺-X⁵⁺ and M³⁺-X⁴⁺) compositions as potential solid electrolytes, *Chem. Mater.* 24 (2012) 1357–1364.
- [44] R. Jaleem, M. Nakayama, T. Kasuga, An efficient rule-based screening approach for discovering fast lithium ion conductors using density functional theory and artificial neural networks, *J. Mater. Chem. A* 2 (2014) 720–734.
- [45] P. Friederich, F. Häse, J. Proppe, A. Aspuru-Guzik, Machine-learned potentials for next-generation matter simulations, *Nat. Mater.* 20 (2021) 750–761.
- [46] V. Lacivita, N. Artrith, G. Ceder, Structural and compositional factors that control the Li-ion conductivity in LiPON electrolytes, *Chem. Mater.* 30 (2018) 7077–7090.
- [47] W. Li, Y. Ando, E. Minamitani, S. Watanabe, Study of Li atom diffusion in amorphous Li₃PO₄ with neural network potential, *J. Chem. Phys.* 147 (2017) 214106.
- [48] K. Miwa, R. Asahi, Molecular dynamics simulations of lithium superionic conductor Li₁₀GeP₂S₁₂ using a machine learning potential, *Solid State Ion.* 361 (2021).

- [49] J. Huang, L. Zhang, H. Wang, J. Zhao, J. Cheng, W. E. Deep potential generation scheme and simulation protocol for the $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ -type superionic conductors, *J. Chem. Phys.* 154 (2021) 094703.
- [50] A.D. Sendek, Q. Yang, E.D. Cubuk, K.-A.N. Duerloo, Y. Cui, E.J. Reed, Holistic computational structure screening of more than 12000 candidates for solid lithium-ion conductor materials, *Energy Environ. Sci.* 10 (2017) 306–320.
- [51] H.Y. Guo, Q. Wang, A. Urban, N. Artrith, Artificial intelligence-aided mapping of the structure-composition-conductivity relationships of glass-ceramic lithium thiophosphate electrolytes, *Chem. Mater.* 34 (2022) 6702–6712.
- [52] Y. Zhang, X. He, Z. Chen, Q. Bai, A.M. Nolan, C.A. Roberts, D. Banerjee, T. Matsunaga, Y. Mo, C. Ling, Unsupervised discovery of solid-state lithium ion conductors, *Nat. Commun.* 10 (2019).
- [53] K. Tao, Z. Wang, Y. Han, J. Li, Rapid discovery of inorganic-organic solid composite electrolytes by unsupervised learning, *Chem. Eng. J.* 454 (2023) 140151.
- [54] G. Bradford, J. Lopez, J. Ruza, M.A. Stolberg, R. Osterude, J.A. Johnson, R. Gomez-Bombarelli, Y. Shao-Horn, Chemistry-informed machine learning for polymer electrolyte discovery, *ACS Cent. Sci.* 9 (2023) 206–216.
- [55] Z. Ahmad, T. Xie, C. Maheshwari, J.C. Grossman, V. Viswanathan, Machine learning enabled computational screening of inorganic solid electrolytes for suppression of dendrite formation in lithium metal anodes, *ACS Cent. Sci.* 4 (2018) 996–1006.
- [56] E.D. Cubuk, A.D. Sendek, E.J. Reed, Screening billions of candidates for solid lithium-ion conductors: a transfer learning approach for small data, *J. Chem. Phys.* 150 (2019) 214701.
- [57] S. Ibrahim, M.R. Johan, Conductivity, Thermal and neural network model nano-composite solid polymer electrolyte system (PEO-LiPF₆-EC-CNT), *Int. J. Electrochem. Sci.* 6 (2011) 5565–5587.
- [58] M.S. Beal, B.E. Hayden, T. Le Gall, C.E. Lee, X.J. Lu, M. Mirsaneh, C. Mormiche, D. Pasero, D.C.A. Smith, A. Weld, C. Yada, S. Yokoishi, High throughput methodology for synthesis, screening, and optimization of solid state lithium ion electrolytes, *ACS Comb. Sci.* 13 (2011) 375–381.
- [59] Y.-T. Chen, M. Duquesnoy, D.H.S. Tan, J.-M. Daux, H. Yang, G. Deysher, P. Ridley, A.A. Franco, Y.S. Meng, Z. Chen, Fabrication of high-quality thin solid-state electrolyte films assisted by machine learning, *ACS Energy Lett.* (2021) 1639–1648.
- [60] C.W. Luo, Q.L. Shuai, G.Q. Zhao, M.Y. Zhang, B. Wu, X.L. Fu, Y.J. Sun, Y. Wang, Q.S. Hua, Insights into the Effects of Co-Regulated Factors on $\text{Li}_{1.3}\text{Al}_{0.5}\text{Ti}_{1.7}(\text{PO}_4)_3$ Solid electrolyte preparation: sources, calcination temperatures, and sintering temperatures, *ACS Appl. Mater. Interfaces* 15 (2023) 48110–48121.
- [61] J.K. Eckhardt, S. Kremer, T. Fuchs, P. Minnmann, J. Schubert, S. Burkhardt, M.T. Elm, P.J. Klar, C. Heiliger, J. Janek, Influence of microstructure on the material properties of LLZO ceramics derived by impedance spectroscopy and brick layer model analysis, *ACS Appl. Mater. Interfaces* 15 (2023) 47260–47277.
- [62] G. Yang, M.L. Lehmann, S. Zhao, B. Li, S. Ge, P. Cao, F.M. Delnick, A.P. Sokolov, T. Saito, J. Nanda, Anomalous high elastic modulus of a poly (ethylene oxide)-based composite electrolyte, *Energy Storage Mater.* 35 (2021) 431–442.
- [63] W. Chen, Y. Li, D. Feng, C. Lv, H. Li, S. Zhou, Q. Jiang, J. Yang, Z. Gao, Y. He, J. Luo, Recent progress of theoretical research on inorganic solid state electrolytes for Li metal batteries, *J. Power Sources* 561 (2023) 232720.
- [64] J.Z.A.Y. Zhuoyuan Zheng, Computational approach inspired advancements of solid-state electrolytes for lithium secondary batteries: from first-principles to machine learning, *Chem. Soc. Rev.* (2024) 3134.
- [65] D. Diddens, W.A. Appiah, Y. Mabrouk, A. Heuer, T. Vegge, A. Bhowmik, Modeling the solid electrolyte interphase: machine learning as a game changer? *Adv. Mater. Interfaces* 9 (2022).
- [66] L. Himanen, A. Geurts, A.S. Foster, P. Rinke, Data-driven materials science: status, challenges, and perspectives, *Adv. Sci.* 6 (2019) 1900808.
- [67] M. Hellenbrandt, The Inorganic Crystal structure database (ICSD)—present and future, *Crystallogr. Rev.* 10 (2004) 17–22.
- [68] S. Gražulis, D. Chateigner, R.T. Downs, A.F.T. Yokochi, M. Quirós, L. Lutterotti, E. Manakova, J. Butkus, P. Moeck, A. Le Bail, Crystallography open database – an open-access collection of crystal structures, *J. Appl. Crystallogr.* 42 (2009) 726–729.
- [69] C.R. Groom, I.J. Bruno, M.P. Lightfoot, S.C. Ward, The Cambridge structural database, *Acta Crystallogr. Sect. B Struct. Sci. Cryst. Eng. Mater.* 72 (2016) 171–179.
- [70] M.Z. Mayers, J.W. Kaminski, T.F. Miller, Suppression of dendrite formation via pulse charging in rechargeable lithium metal batteries, *J. Phys. Chem. C* 116 (2012) 26214–26221.
- [71] J.E. Saal, S. Kirklin, M. Aykol, B. Meredig, C. Wolverton, Materials design and discovery with high-throughput density functional theory: the open quantum materials Database (OQMD), *JOM* 65 (2013) 1501–1509.
- [72] C. Draxl, M. Scheffler, The NOMAD laboratory: from data sharing to artificial intelligence, *J. Phys.: Mater.* 2 (2019) 036001.
- [73] L. Talirz, S. Kumbhar, E. Passaro, A.V. Yakutovich, V. Granata, F. Gargiulo, M. Borelli, M. Uhrin, S.P. Huber, S. Zoupanos, C.S. Adorf, C.W. Andersen, O. Schütt, C.A. Pignedoli, D. Passerone, J. Vandevondele, T.C. Schulthess, B. Smit, G. Pizzi, N. Marzari, Materials Cloud, a platform for open computational science, *Sci. Data* 7 (2020).
- [74] J. Schmidt, M.R.G. Marques, S. Botti, M.A.L. Marques, Recent advances and applications of machine learning in solid-state materials science, *NPJ Comput. Mater.* 5 (2019).
- [75] R. Batra, L. Song, R. Ramprasad, Emerging materials intelligence ecosystems propelled by machine learning, *Nat. Rev. Mater.* 6 (2020) 655–678.
- [76] L. Zhang, B. He, Q. Zhao, Z. Zou, S. Chi, P. Mi, A. Ye, Y. Li, D. Wang, M. Avdeev, S. Adams, S. Shi, A database of ionic transport characteristics for over 29 000 inorganic compounds, *Adv. Funct. Mater.* 30 (2020) 2003087.
- [77] Z.L. Wang, J. Gao, K.H. Tao, Y.Q. Han, A. Chen, J.J. Li, IonML: a physically inspired machine learning platform to directed design superionic conductors, *Energy Storage Mater.* 59 (2023).
- [78] C.J. Hargreaves, M.W. Gaultois, L.M. Daniels, E.J. Watts, V.A. Kurlin, M. Moran, Y. Dang, R. Morris, A. Morscher, K. Thompson, M.A. Wright, B. Prasad, F. Blanc, C.M. Collins, C.A. Crawford, B.B. Duff, J. Evans, J. Gamon, G. Han, B.T. Leube, H. Niu, A.J. Perez, A. Robinson, O. Rogan, P.M. Sharp, E. Shoko, M. Sonni, W.J. Thomas, A. Vasylenko, L. Wang, M.J. Rosseinsky, M.S. Dyer, A database of experimentally measured lithium solid electrolyte conductivities evaluated with machine learning, *NPJ Comput. Mater.* 9 (2023) 9–14.
- [79] H. Guo, M.R. Carbone, C. Cao, J. Qu, Y. Du, S.-M. Bak, C. Weiland, F. Wang, S. Yoo, N. Artrith, A. Urban, D. Lu, Simulated sulfur K-edge X-ray absorption spectroscopy database of lithium thiophosphate solid electrolytes, *Sci. Data* 10 (2023).
- [80] D. Fourches, E. Muratov, A. Tropsha, Trust, But Verify: On the importance of chemical structure curation in cheminformatics and QSAR modeling research, *J. Chem. Inf. Model.* 50 (2010) 1189–1204.
- [81] D. Fourches, E. Muratov, A. Tropsha, Trust, but Verify II: a practical guide to chemogenomics data curation, *J. Chem. Inf. Model.* 56 (2016) 1243–1252.
- [82] P. Raccuglia, K.C. Elbert, P.D.F. Adler, C. Falk, M.B. Wenny, A. Molloy, M. Zeller, S.A. Friedler, J. Schrier, A.J. Norquist, Machine-learning-assisted materials discovery using failed experiments, *Nature* 533 (2016) 73–76.
- [83] P. Tremouilhac, A. Nguyen, Y.-C. Huang, S. Kotov, D.S. Lütjohann, F. Hübsch, N. Jung, S. Bräse, Chemotion ELN: an open source electronic lab notebook for chemists in academia, *J. Cheminform.* 9 (2017).
- [84] H. El-Bousidy, T. Lombardo, E.N. Primo, M. Duquesnoy, M. Morcrette, P. Johansson, P. Simon, A. Grimaud, A.A. Franco, What Can text mining tell us about lithium-ion battery researchers' habits? *Batter. Supercaps* 4 (2021) 758–766.
- [85] R. Mahbub, K. Huang, Z. Jensen, Z.D. Hood, J.L.M. Rupp, E.A. Olivetti, Text mining for processing conditions of solid-state battery electrolytes, *Electrochem. Commun.* 121 (2020) 106860.
- [86] M.D. Wilkinson, M. Dumontier, I.J. Aalbersberg, G. Appleton, M. Axton, A. Baak, N. Blomberg, J.-W. Boiten, L.B. Da Silva Santos, P.E. Bourne, J. Bouwman, A.J. Brookes, T. Clark, M. Crosas, I. Dillo, O. Dumon, S. Edmunds, C.T. Evelo, R. Finkers, A. Gonzalez-Beltran, A.J.G. Gray, P. Groth, C. Goble, J.S. Grethe, J. Heringa, P.A.C. 'T. Hoen, R. Hoof, T. Kuhn, R. Kok, J. Kok, S.J. Lusher, M.E. Martone, A. Mons, A.L. Packer, B. Persson, P. Rocca-Serra, M. Roos, R. Van Schaik, S.-A. Sansone, E. Schultes, T. Sengstag, T. Slater, G. Strawn, M.A. Swertz, M. Thompson, J. Van Der Lei, E. Van Mulligen, J. Velterop, A. Waagmeester, P. Wittenburg, K. Wolstencroft, J. Zhao, B. Mons, The FAIR guiding principles for scientific data management and stewardship, *Sci. Data* 3 (2016) 160018.
- [87] L.M. Ghiringhelli, J. Vybiral, S.V. Levchenko, C. Draxl, M. Scheffler, Big data of materials science: Critical Role of the Descriptor, *Phys. Rev. Lett.* 114 (2015).
- [88] Y.Y. Lin, A.X. Bin Yong, W.J. Gustafson, C.N. Reedy, E. Ertekin, J.A. Krogstad, N.H. Perry, Toward design of cation transport in solid-state battery electrolytes: Structure-dynamics relationships, *Curr. Opin. Solid State Mater. Sci.* 24 (2020).
- [89] Y. Li, T. Li, H. Liu, Recent advances in feature selection and its applications, *Knowl. Inf. Syst.* 53 (2017) 551–577.
- [90] L. Himanen, M.O.J. Jäger, E.V. Morooka, F.F. Canova, Y.S. Ranawat, D.Z. Gao, P. Rinke, A.S. Foster, D.Scribe: library of descriptors for machine learning in materials science, *Comput. Phys. Commun.* 247 (2020).
- [91] S. Li, Y. Liu, D. Chen, Y. Jiang, Z. Nie, F. Pan, Encoding the atomic structure for machine learning in materials science, *WIREs Comput. Mol. Sci.* 12 (2022).
- [92] W. Jia, M. Sun, J. Lian, S. Hou, Feature dimensionality reduction: a review, *Complex Intell. Syst.* 8 (2022) 2663–2693.
- [93] O. Isayev, C. Oses, C. Toher, E. Gossett, S. Curtarolo, A. Tropsha, Tropsha, Universal fragment descriptors for predicting properties of inorganic crystals, *Nat. Commun.* 8 (2017) 15679.
- [94] M. Rupp, A. Tkatchenko, K.-R. Müller, O.A. Von Lilienfeld, Fast and accurate modeling of molecular atomization energies with machine learning, *Phys. Rev. Lett.* 108 (2012).
- [95] J. Behler, Atom-centered symmetry functions for constructing high-dimensional neural network potentials, *J. Chem. Phys.* 134 (2011) 074106.
- [96] A.P. Bartók, R. Kondor, G. Csányi, On representing chemical environments, *Phys. Rev. B* 87 (2013).
- [97] T. Xie, J.C. Grossman, Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties, *Phys. Rev. Lett.* 120 (2018).
- [98] T. Zhou, Z. Song, K. Sundmacher, Big data creates new opportunities for materials research: a review on methods and applications of machine learning for materials design, *Engineering* 5 (2019) 1017–1026.
- [99] R. Ramprasad, R. Batra, G. Pilania, A. Mannodi-Kanakkithodi, C. Kim, Machine learning in materials informatics: recent applications and prospects, *NPJ Comput. Mater.* 3 (2017).
- [100] R. Pollice, G.D. Gomes, M. Aldeghi, R.J. Hickman, M. Krenn, C. Lavigne, M. Lindner-D'Addario, A. Nigam, C.T. Ser, Z.P. Yao, A. Aspuru-Guzik, Data-driven strategies for accelerated materials design, *Acc. Chem. Res.* 54 (2021) 849–860.
- [101] K. Choudhary, B. Decost, C. Chen, A. Jain, F. Tavazza, R. Cohn, C.W. Park, A. Choudhary, A. Agrawal, S.J.L. Billinge, E. Holm, S.P. Ong, C. Wolverton, Recent advances and applications of deep learning methods in materials science, *NPJ Comput. Mater.* 8 (2022).
- [102] I. Portugal, P. Alencar, D. Cowan, The use of machine learning algorithms in recommender systems: a systematic review, *Expert Syst. Appl.* 97 (2018) 205–227.
- [103] H. Borchani, G. Varando, C. Bielza, P. Larrañaga, A survey on multi-output regression, *WIREs Data Min. Knowl. Discov.* 5 (2015) 216–233.
- [104] O. Sagi, L. Rokach, Ensemble Learning: A Survey, *WIREs Data Min. Knowl. Discov.* 8 (2018) e1249.

- [105] S. Dong, P. Wang, K. Abbas, A survey on deep learning and its applications, *Comput. Sci. Rev.* 40 (2021).
- [106] Q.T. Sun, Y. Xiang, Y. Liu, L. Xu, T.L. Leng, Y.F. Ye, A. Fortunelli, W.A. Goddard, T. Cheng, Machine learning predicts the x-ray photoelectron spectroscopy of the solid electrolyte interface of lithium metal battery, *J. Phys. Chem. Lett.* 13 (2022) 8047–8054.
- [107] J. Kim, S. Kang, K.Y.M. Min, Screening platform for promising Na superionic conductors for Na-ion solid-state electrolytes, *ACS Appl. Mater. Interfaces* 15 (2023) 41417–41425.
- [108] S. Kang, M. Kim, K. Min, Discovery of superionic solid-state electrolyte for Li-Ion batteries via machine learning, *J. Phys. Chem. C* 127 (2023) 19335–19343.
- [109] I. Mandal, S. Mannan, L. Wondraczek, N.N. Gosvami, A.R. Allu, N.M.A. Krishnan, Machine Learning-assisted design of Na-Ion-conducting glasses, *J. Phys. Chem. C* 127 (2023) 14636–14644.
- [110] P.C. Xu, T. Lu, X.B. Ji, M.J. Li, W.C. Lu, Machine Learning combined with weighted voting regression and proactive searching progress to discover $\text{ABO}_{3-\delta}$ perovskites with high oxide ionic conductivity, *J. Phys. Chem. C* 127 (2023) 17096–17108.
- [111] J. Sun, S. Kang, J. Kim, K. Min, Accelerated discovery of novel garnet-type solid-state electrolyte candidates via machine learning, *ACS Appl. Mater. Interfaces* 15 (2023) 5049–5057.
- [112] C. Chen, Y. Zuo, W. Ye, X. Li, Z. Deng, S.P. Ong, A critical review of machine learning of energy materials, *Adv. Energy Mater.* 10 (2020) 1903242.
- [113] T. Lombardo, M. Duquesnoy, H. El-Bouysidi, F. Årén, A. Gallo-Bueno, P.B. Jorgensen, A. Bhowmik, A. Demortiere, E. Ayerbe, F. Alcaide, M. Reynaud, J. Carrasco, A. Grimaud, C. Zhang, T. Vegge, P. Johansson, A.A. Franco, Artificial intelligence applied to battery research: hype or reality? *Chem. Rev.* 122 (2022) 10899–10969.
- [114] Y. Liu, B.R. Guo, X.X. Zou, Y.J. Li, S.Q. Shi, Machine learning assisted materials design and discovery for rechargeable batteries, *Energy Storage Mater.* 31 (2020) 434–450.
- [115] Y. Liu, J.M. Wu, M. Avdeev, S.Q. Shi, Multi-Layer feature selection incorporating weighted score-based expert knowledge toward modeling materials with targeted properties, *Adv. Theory Simul.* 3 (2020) 1900215.
- [116] P. Xu, X. Ji, M. Li, W. Lu, Small data machine learning in materials science, *NPJ Comput. Mater.* 9 (2023).
- [117] C. Leng, Z. Tang, Y.G. Zhou, Z. Tian, W.Q. Huang, J. Liu, K.Q. Li, K.L. Li, Fifth paradigm in science: a case study of an intelligence-driven material design, *Engineering* 24 (2023) 126–137.
- [118] F. Li, X. Cheng, L. Lu, Y. Yin, J. Luo, G. Lu, Y. Meng, H. Mo, T. Tian, J. Yang, W. Wen, Z. Liu, G. Zhang, C. Shang, H. Yao, Stable all-solid-state lithium metal batteries enabled by machine learning simulation designed halide electrolytes, *Nano Lett.* 22 (2022) 2461–2469.
- [119] J. Li, M. Zhou, H.H. Wu, L. Wang, J. Zhang, N. Wu, K. Pan, G. Liu, Y. Zhang, J. Han, X. Liu, X. Chen, J. Wan, Q. Zhang, Machine learning-assisted property prediction of solid-state electrolyte, *Adv. Energy Mater.* 14 (2024).
- [120] OpenAI, “GPT-4 Technical Report,” vol. 4, pp. 1–100, 2023, [Online]. Available: <http://arxiv.org/abs/2303.08774>.
- [121] N.J. Szymanski, B. Rendy, Y. Fei, R.E. Kumar, T. He, D. Milsted, M.J. Mcdermott, M. Gallant, E.D. Cubuk, A. Merchant, H. Kim, A. Jain, C.J. Bartel, K. Persson, Y. Zeng, G. Ceder, An autonomous laboratory for the accelerated synthesis of novel materials, *Nature* 624 (2023) 86–91.